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Differential expression of jackfruit-lectin-specific glycoconjugates in metastatic adenocarcinoma and reactive mesothelial cells – a diagnostic aid in effusion cytology

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Abstract Distinguishing reactive mesothelial cells from adenocarcinoma cells in serous effusions on the basis of morphological criteria alone is often difficult. Interest has therefore been focused on identifying reliable methods to supplement the conventional cytological techniques. Plant lectins have been reported as diagnostic markers for malignant cells. We studied 51 aspirated samples of benign and malignant effusions using horseradish-peroxidase-conjugated jackfruit lectin. No significant difference was observed between the cells of pleural and peritoneal fluids. The reactively proliferated mesothelial cells of benign effusions showed a predominance of mild staining while moderate and intense staining was predominant in malignant effusions. Intense and irregular lectin binding was observed in macrophages irrespective of the cause of effusion. The lectin staining method therefore appears to have some clinical significance as an additional diagnostic aid for use in effusion cytology.

Key words Effusions · Jackfruit lectin · Diagnostic cytology · Malignancy

Introduction

The presence of ascitis or pleural effusion in patients without any evidence of infection or any other metabolic

aetiology often results in a diagnostic dilemma. Malignant effusion in many of the cancer patients, particularly in ovarian cancer with frequent spread of tumour into the peritoneal cavity, is a common complication. Cytological examination of ascitis or pleural fluid often helps the diagnostic distinction between benign and malignant effusions in most cases, but it fails to allow a definite diagnosis in about 15% of cases (Robinson and Royston 1993). The changing therapy, infection and the different laboratory processing techniques may cause the benign reactive cells to appear so atypical that determining whether the cells are benign or malignant becomes difficult. The conventional histochemical stains like periodic acid/Schiff (PAS) diastase are not reliable discriminants (Bollinger et al. 1989). One alternative approach to supplement cytological evaluation is immunocytochemistry, but it is still not known which antibody is the most suitable for this purpose, as normal and reactively proliferated mesothelial cells also express most of the antigens seen in malignant cells (Mezger et al. 1992). Recently we have reported the significance of the silver-staining nucleolar organiser region count to supplement the cytological evaluation of effusions (Sujathan et al. 1996). In the present study we have analysed the potential of a lectin from jackfruit to distinguish malignant cells from benign reactive cells in effusions.

Lectins are proteins or glycoproteins extracted predominantly from plants and have been used in many areas of diagnostic investigation, especially those related to changes in the expression of membrane and cytoplasmic glycoconjugates (Walker 1989; Letham and Brooks 1987). As neoplastic transformation of cells is associated with altered cell-surface glycoconjugates, a large number of studies have been performed to differentiate malignant cells from benign cells using different lectins. *N*-Acetyl-D-galactosamine-specific jackfruit lectin has been reported to be useful in the diagnosis of various tumours (Vijayan et al. 1982; Vijayakumar et al. 1987).

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Materials and methods

Alcohol-fixed smears from 51 serous effusions were used for this study. A total of 20 samples were cytologically positive with histologically diagnosed primary adenocarcinoma, which includes carcinoma of the ovary (10), carcinoma of the breast (5), carcinoma of the lung (3) and carcinoma of the stomach (2); 31 samples were cytologically negative and a benign cause of the effusion had been identified. Among these, 20 samples were obtained from patients with pulmonary tuberculosis referred from the Sanatorium for Chest Disease and 11 from patients with liver cirrhosis referred from the Medical Gastroenterology Department of the Medical College hospital. The jackfruit lectin was isolated from the seeds of jackfruit (*Artocarpus integrifolia*), purified and conjugated to horseradish peroxidase type IV according to the technique described elsewhere (Vijayakumar et al. 1987). Smears were treated with 0.3% hydrogen peroxide in methanol to block the endogenous peroxidase activity and were rinsed in distilled water followed by two changes of phosphate-buffered saline (PBS, pH 7.4). The smears were then incubated with horseradish-peroxidase-conjugated jackfruit lectin in a moist chamber for 30 min and again rinsed in three changes of PBS. The lectin-bound areas were visualized by treatment with a solution of 30 mg/100 mg diaminobenzidine dihydrochloride in PBS containing 40 ml hydrogen peroxide for 5 min (Nakane and Kawaoi 1974). Finally the smears were counterstained with Harris haematoxylin and mounted in Dextrin polyesterin xylene (DPX). The smears were observed under a light microscope. The intensity of staining was assessed on an arbitrary scale as mild (+), moderate (++) and intense (+++). In malignant effusions the malignant cells and in benign effusions the mesothelial cells were studied; 100 cells were assessed in each sample.

Statistical analysis

The percentage of cells showing different grades of staining was noted in each sample and tabulated. Univariate statistics, such as the mean, standard deviation, standard error, median, minimum/maximum, variance and 95% confidence interval (CI) of the mean, were calculated for each staining grade in both benign and malignant lesions. Student's *t*-tests were applied to the mean values of different staining grades, comparing benign and malignant lesions to assess the statistical significance.

Results

No significant difference in staining pattern was observed between the cells of pleural and peritoneal fluids. The qualitative difference in staining patterns of benign and malignant effusions is shown in Fig. 1. The staining pattern was found to be uniform in both benign and malignant cells. A higher percentage of cells with mild staining was seen in benign effusions (83.97 ± 16.58) (Fig. 2D) while the percentage of cells with moderate and intense staining patterns was higher in malignant effusions ($51.13 \pm 27.73\%$ and $33.78 \pm 31.99\%$ respectively) (Fig. 2F). In macrophages, the lectin binding was as intense as in the malignant cells, but staining was irregular and somewhat granular in pattern (Fig. 2E). One of the cytologically benign samples showed a higher percentage of cells with intense staining. A review of the Papanicolaou and May-Grünwald-Giemsa-stained smears of this sample revealed the presence of a large number of macrophages. The lectin-stained smear was reassessed and the intensely stained cells were differentiated as macrophages.

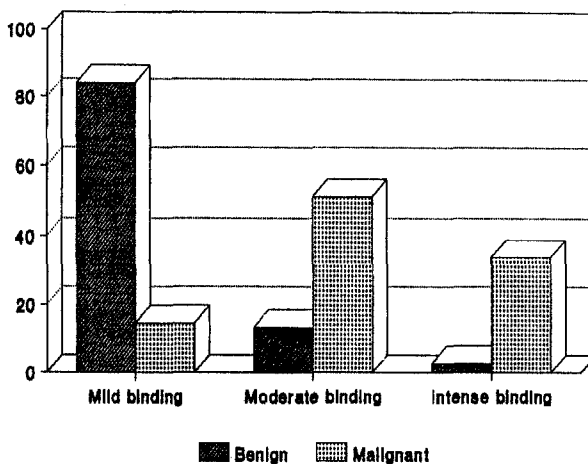


Fig. 1 Bar graph showing the percentage of cells having different grades of lectin binding in benign and malignant effusions

Statistical analysis

Univariate statistics on the percentages of cells having different grades of staining in benign and malignant effusions are shown in Table 1. The malignant lesions were more heterogeneous than the benign lesions in all staining grades as shown by the variance and CI ranges. The *t*-tests showed significant differences between benign and malignant lesions for all grades of staining, as shown in Table 2. The most prominent significant difference was observed in the percentages of cells having a mild grade of staining.

Discussion

The cytomorphological changes during the process of malignant transformation are either preceded or accompanied by biochemical alterations in the neoplastic cells. Among these alterations, expression of cell-surface glycoprotein plays a vital role, since this is mainly involved in signal transduction phenomena (Gabius 1988). The ability of plant lectins to make such alterations visible and thereby to distinguish normal and neoplastic cells was reported as early as 1963 (Aub et al. 1963). Since then quite a large number of studies have used different types of lectins as histochemical probes to demonstrate alterations in the terminal carbohydrate sequence of cell membrane glycoconjugates in a variety of neoplastic lesions, including those of the breast (Louis et al. 1983; Stanley et al. 1986), salivary gland (Daley et al. 1985), colon (Rhodes et al. 1986), prostate (Soderstrom 1987), endometrium (Yen 1986) and ovary (Teshima et al. 1984). Studies using concanavalin A and peanut agglutinin were found to differentiate benign and malignant cells in tissue sections by the difference in membrane and cytoplasmic staining pattern (Soderstrom 1987; Kannan et al. 1993). Vijayakumar et al. (1987) have shown that jackfruit lectin could bind

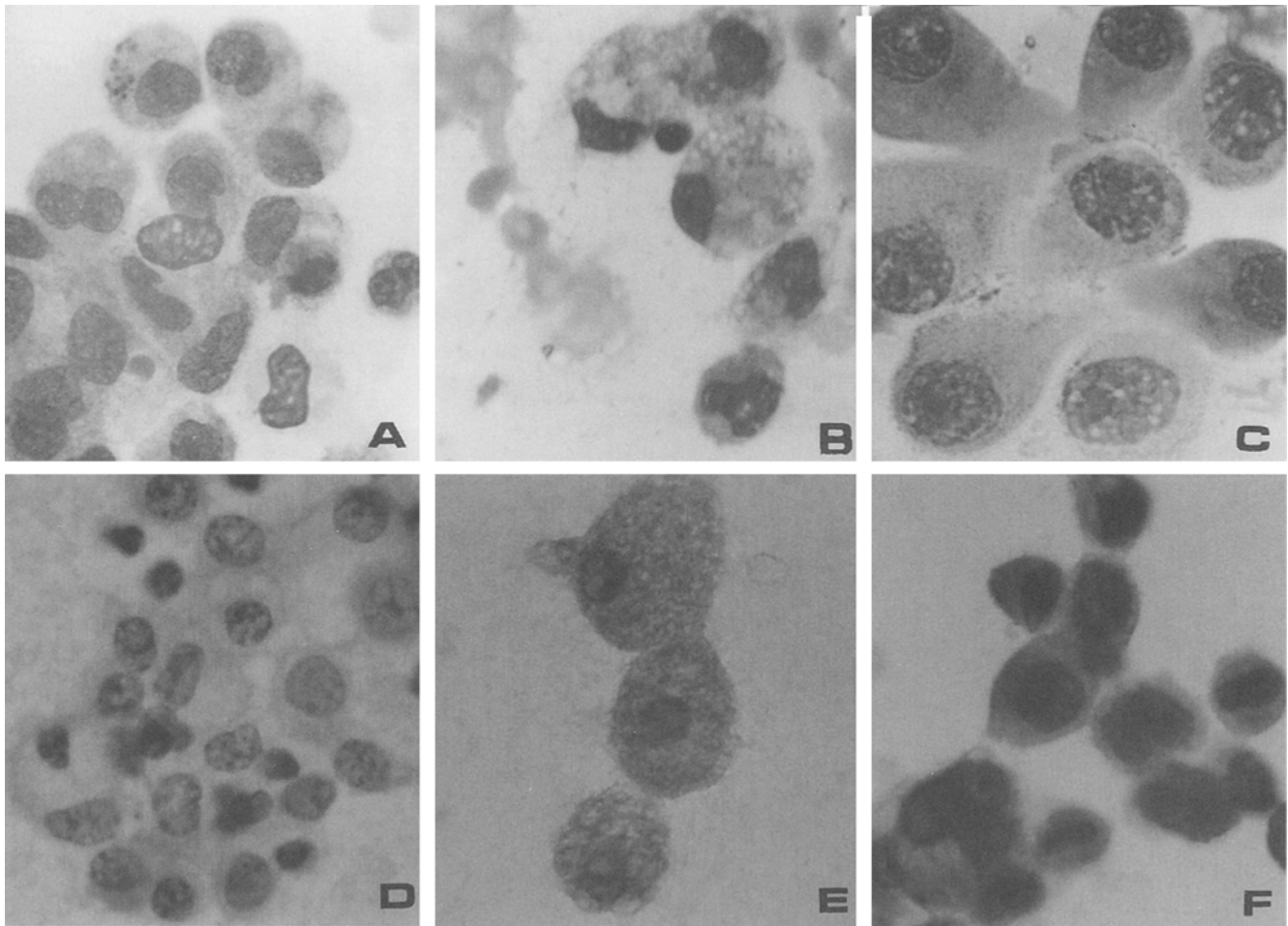


Fig. 2 May-Grünwald-Giemsa (MGG) staining and the corresponding jackfruit-lectin-stained cells. **A** Reactive mesothelial cells of benign effusion (MGG $\times 1000$). **B** Macrophages in benign effusion (MGG $\times 1000$). **C** Adenocarcinoma cells of malignant effusion (MGG $\times 1000$). **D** Mild grade of jackfruit lectin binding pattern in reactive mesothelial cells of benign effusion (1000 $\times$). **E** Intense irregular granular staining pattern of macrophages seen in effusion (1000 $\times$). **F** Intense uniform jackfruit lectin binding in adenocarcinoma cells of malignant effusion (1000 $\times$)

astrocytoma cells in culture. Jackfruit lectin has been reported to be a diagnostic marker for carcinoma of the cervix and malignant and premalignant lesions of the oral cavity (Vijayan et al. 1982, 1987). The intense membrane-binding pattern of this lectin was found to differentiate neoplastic cells of the breast from normal and hyperplastic cells in formalin-fixed tissues (Remani et al. 1989). Similar results have been reported in the thyroid and respiratory tract (Vijayakumar et al. 1992; Pillai et al. 1992). But this technique has rarely been used in cytology. The available report on the use of lectins in effusion cytology revealed that the lectins are useful in distinguishing malignant cells from benign cells and unsuccessful in determining the tumour cell origin (Abramenko et al. 1991). Jackfruit lectin has been shown to have some value in assessing the malignant potential in cervical lesions and to be useful in identifying the malignant cells in sputum (Pillai et al. 1992, 1994).

The present study substantiates the potential use of lectins in diagnostic cytopathology. It is difficult to differentiate membrane and cytoplasmic staining distinctly as seen in tissue sections owing to the presence of plasma membrane all over the cell. Moreover lectins show considerable affinity to cell-surface glycoproteins. The reactively proliferated mesothelial cells in benign effusions showed mostly mild staining whereas moderate to intense binding

was observed in all the samples of malignancies. This difference in binding may be due to the fact that jackfruit lectin may be able to identify incomplete non-sialated forms of membrane glycoconjugates expressed more at the surface of malignant cells, as reported earlier by Ross et al. (1980). However, the macrophages present in reactive effusions also showed much intense staining but could be easily differentiated from malignant cells by the irregular granular staining pattern. A similar staining pattern of pulmonary macrophages has been reported in the respiratory tract (Pillai et al. 1992).

The difficulty in the cytodagnosis of effusions is mainly in differentiating adenocarcinoma cells from mesothelial cells with reactive changes. The present study clearly demonstrated the value of lectins to distinguish malignant cells from benign cells. It is a relatively new field of

Table 1 Univariate statistics of jackfruit-lectin-binding patterns in serous effusion cells

Univariate statistics	Lectin-binding pattern (% of cells)					
	Mild		Moderate		Intense	
	Benign	Malignant	Benign	Malignant	Benign	Malignant
Mean	84	15	13	51	3	34
Standard deviation	17	22	8	28	13	32
Standard error	3	5	1	6	2	7
Median	89	7	11	60	0	26
Minimum	6	0	4	2	0	0
Maximum	96	97	36	98	73	98
Range	90	97	32	96	73	98
Variance	275	490	62	769	172	1024
Confidence interval	78–90	3–24	10–16	39–66	–2–7	18–49

Table 2 *t*-test of jackfruit-lectin-binding patterns in serous effusion cells: means $\pm$ SD

Cytology (n)	Lectin-binding pattern (% of cells)		
	Mild	Moderate	Intense
Benign (31)	83.97 $\pm$ 16.58	13.32 $\pm$ 7.89	2.71 $\pm$ 13.12
Malignant (20)	14.65 $\pm$ 22.13	51.13 $\pm$ 27.73	33.78 $\pm$ 31.99
<i>t</i>	12.62	6.35	4.39
Significance	<0.001	<0.001	<0.001

application in cytopathology. The technique is very simple and inexpensive compared to any other immunocytochemical methods. Hence the present study suggests that the lectin staining method may have some value as an additional diagnostic tool for use in ascitic and pleural fluid samples when cytological diagnosis is difficult.

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